ApexSphere Infrastructure Risk and Resilience Assessment

Prepared by Engineering & Risk · Scope: ATL-1 · Classification: Internal (fictional)

1. Risk summary

The overall risk rating for ATL-1 power infrastructure is High. UPS Bank B batteries are at 82% of their rated service life and were the root cause of the February 11 failover.

Hall 3 is served by a single utility feed, so a utility-side fault would leave it on generator power with no redundant grid path. Combined with the end-of-life Chiller Plant 2, these issues create correlated failure modes during summer peak load.

Finding	Rating
UPS Bank B battery life	High
Hall 3 single utility feed	High
Chiller Plant 2 end of life	Medium-High

2. Remediation plan

Engineering recommends three projects: replace the UPS Bank B batteries ($6.2 million, about 4 months); add a second utility feed to Hall 3 ($21.0 million, about 14 months); and replace Chiller Plant 2 ($9.5 million, about 9 months).

The total engineering estimate is $36.7 million before contingency. The battery replacement can begin immediately; the second feed depends on the same utility engineering team as any new substation work.

3. Exposure if remediation is deferred

Engineering estimates an 18% probability of a site-wide outage at ATL-1 over the next 24 months if remediation is deferred, falling to 4% once all three projects are complete.

The estimated cost of a site-wide outage is $2.1 million per hour, including SLA credits and emergency response, before any reputational impact. Engineering advises sequencing remediation ahead of, or in parallel with, any major expansion that draws on the same utility and construction resources.